CSE499B – Section 15 – Group 03
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CityPulse: Intelligent Parking & Traffic Control Platform

Weekly Progress Report – Update 3 (Sprint 3: Cameras, Traffic Forecast and Navigation)

I. Sprint Overview

This update covers 28 October – 4 November 2026. The backend added camera-by-camera lot setup, vehicle prices, a traffic jam forecast, and a first end-to-end test of all AI models on Dhaka images. The web portal now stays reliable when the server is asleep or a token expires. The driver app got a rebuilt map with fast search and in-app navigation, the booking rules were tightened, and a YOLOv12 paper was reproduced for plate detection.

II. AI & Backend Progress

- **Cameras module:** lets an owner register a parking lot one camera at a time, together with the slots that camera sees. A failure during setup loses at most one camera, and cameras can be added later.
  - *Add a camera:* saves the camera's source (RTSP stream or uploaded video), frame size and its slots; every slot box must be inside the frame, must not overlap another, and needs a unique label in the lot.
  - *Edit a camera:* changes the camera and its slots; known slot labels are updated and new ones added.
  - *List cameras:* returns all cameras of a lot with their slots, boxes and prices, so the owner can reopen the setup.
  - *Slot boxes:* drawn in pixels, stored as fractions of the frame, so they stay correct when the resolution changes.
- **Rates module:** prices a lot by vehicle type instead of slot by slot.
  - *Set vehicle rate:* hourly price for car, motorcycle, CNG, microbus, SUV, pickup, bicycle or truck; a slot keeps its own price only when it differs.
- **Bookings module:** keeps bookings fair and prevents double booking.
  - *Booking limits:* at most two live bookings per driver, at most 30 days ahead and 24 hours long, five cancellations a day.
  - *No double booking:* no overlapping bookings for a driver or vehicle; all checks run in one locked step, so two drivers cannot take the same slot.
- **Traffic module:** predicts traffic jams around each lot, as the base for offering free slots nearby during a jam.
  - *Jam forecast:* checks four roads into a lot (from 2 km N, E, S, W) with the Google Routes API [1], 30 minutes ahead against the empty road at 3 am; 1.3× slower is *slow*, 1.8× is *jam*.
  - *Result:* Dhaka evening ratios of 1.39–1.66; TomTom has no Dhaka data. Results are cached to stay in the free quota.
- **Database:** new fields for cameras and prices.
  - *Migration:* camera source and media fields, vehicle type on price plans, and bicycle and truck types; applied to the live database.
  - *Testing:* 58 endpoint checks against the real database; the build passes.
- **AI inference tester:** a local page that runs all three AI models on one slot image, the first end-to-end test on Dhaka images.
  - *Occupancy:* MobileNetV2 [2] marks the slot empty or occupied.
  - *Plate detection:* YOLO26n [3] finds the licence plate; weak boxes below 0.4 are dropped.
  - *Plate reading:* YOLO26s reads each character; the plate is printed in Bangla and in English for the database (Table I). Both YOLO models were trained on public Bangla plate datasets [4], [5].

TABLE I
Inference test on Dhaka slot images

Slot	Occupancy	Plate	Plate read (English)
1	occupied, 1.00	0.74	Dhaka Metro 21-8842
6	occupied, 1.00	0.94	Dhaka Metro Ga 24-2955



Fig. 1. AI inference tester: plate found and read on slot 6.

III. Frontend Progress

- **Request pipeline:** the free backend server sleeps when idle and takes about 55 s to wake. Every API call now passes through three layers, each adding one behaviour: a time limit, retries, and the access token. Reads get 60 s and writes 90 s before they are cut off with a clear message ("the server may be waking up") instead of a broken page.
- **Retries:** a read that fails because the server cannot be reached or answers 502, 503 or 504 is retried twice, after

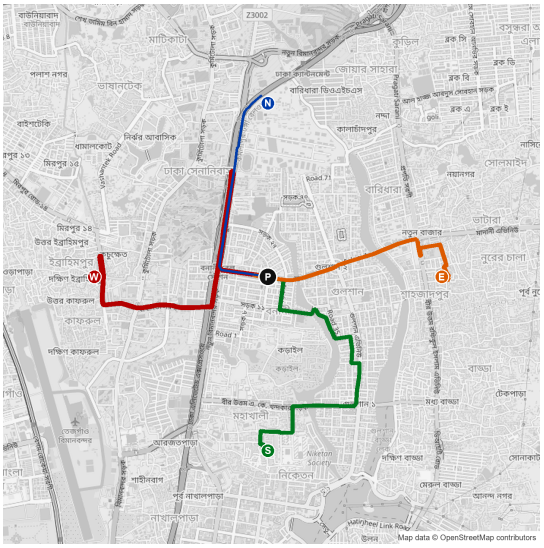


Fig. 2. Traffic module: the four roads into a Gulshan lot (P), checked from 2 km north, east, south and west.

1 s and 3 s. Writes are never retried, so a booking or a lot is never saved twice, and real API errors such as 404 are returned at once.

- **Token refresh:** the API gives a new refresh token on every refresh and rejects the old one, so two requests refreshing at the same moment signed the user out. All requests in a tab now wait on one shared refresh, and a browser lock stops two open tabs from refreshing at the same time.
- **Auth factory:** one setting chooses between the live API and a browser-only demo store for sign-in (useful for UI demos while the server sleeps), and the whole app shares one auth instance, so every request sees the same session.
- **Tests and notes:** the API client has 10 unit tests, covering waking-server retries, writes never retried, no retry on 404, giving up after the last retry, timeouts and cancelling. Before-and-after notes document the changes.
- **Paper review:** HVI-CIDNet [6], a small (1.9 M parameters) low-light image enhancer from CVPR 2025, was reviewed as a pre-processing step for dark parking footage and night plates, the weakest case of the plate models.

IV. Mobile Progress

- **Search:** place search moved to Google Places autocomplete [7] with a free fallback, a short typing delay, request cancelling, a 5-minute cache, and recent searches. Parking lots are matched instantly on the device and listed above place results.
- **Navigation:** routes now come from the Google Routes API [1] with live traffic, with OSRM [8] as a fallback, and the driver is guided inside the app instead of being sent to Google Maps: a turn banner, remaining time and distance, snapping to the road, automatic rerouting when the driver leaves the route, an arrival card, and spoken turn instructions that can be muted.
- **Map:** custom light and dark map styles that follow the app theme, pins that show each lot's number of free slots, a traffic layer, a recenter button, and long-press to drop a pin. Unit tests cover the distance and route-snapping helpers.
- **Bookings:** the booking sheet checks free slots for the chosen start time and duration, booking details and the list were simplified, and in-app notices confirm actions. The new booking rules (Section II) were added with these screens. A new APK was built.
- **Paper reproduction:** YOLOv12 [9], an attention-based detector, was fine-tuned for Bangla plate detection with the authors' code and with Ultralytics, and compared with YOLO11 [3] under the same settings on 1,153 test images (Table II). YOLO11-S is as accurate, about 30% faster, and trains in half the time of the official code; YOLOv12 is best at night with the Ultralytics version (98.4% mAP50). As character readers, YOLO11-N and YOLOv12-N read 83.6% and 82.8% of plates exactly.

TABLE II
Plate detection, YOLOv12 reproduction (1,153 test images)

Model	mAP50	ms (T4)	Train min
YOLOv12-S (official code)	98.6%	21.9	256
YOLO12-S (Ultralytics)	98.9%	20.6	176
YOLO11-S	**99.0%**	**14.8**	**127**

V. Plan for Next Week

- Serve the plate detector and reader from the AI service and store plate reads with the booking's vehicle.
- Backend: camera video upload and streaming, camera delete, a price quote at booking, and jam-time offers from the traffic forecast.
- RTSP stream worker: cut slot crops from live frames for the occupancy model.
- Frontend: owner camera and slot setup on the new API; mobile: jam-time offers and slot selection in the booking flow.

References

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